



Natural and forced convection heat transfer characteristics of single-phase immersion cooling systems for data centers

Yongping Huang^{a,b}, Junlei Ge^b, Yongping Chen^{a,b}, Chengbin Zhang^{a,b,*}

^a Jiangsu Key Laboratory of Micro and Nano Heat Fluid Flow Technology and Energy Application, School of Environmental Science and Engineering, Suzhou University of Science and Technology, Suzhou, Jiangsu 215009, China

^b School of Energy and Environment, Southeast University, Nanjing, Jiangsu 210096, China

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ABSTRACT

Improving power usage effectiveness (PUE) in data centers becomes a key challenge for carbon neutrality and carbon peaking strategies. In this context, this paper investigates the single-phase immersion cooling (SPIC) system that promises to improve PUE. Three-dimensional mathematical models of pump-driven and buoyancy-driven immersion cooling tanks are established and numerically solved. The thermal characteristics, flow resistance, and energy consumption levels of immersion cooling tanks in three operating modes are investigated. Moreover, the roles of volume flow rate, inlet temperature, and device power in immersion cooling characteristics are analyzed. The results indicate that the heat dissipation of buoyancy-driven SPIC systems is less effective than pump-driven ones. Compared to buoyancy-driven SPIC systems, the average temperature of immersion coolants and PUE in pump-driven SPIC systems reduces by 55.5 and 11.6%, respectively. Interestingly, although natural convection in mixed convection operation enhances convective disturbance, it induces temperature stratification that weakens temperature homogeneity. Increases in inlet temperature and device power both worsen the immersion cooling performance. However, increasing inlet temperature improves temperature uniformity and reduces flow resistance losses, while increasing device power has the opposite effect. Besides, the increased flow rate enhances the convective heat transport and temperature uniformity but increases the flow resistance loss.

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1. Introduction

With the booming development of information technologies and explosive growth in data processing needs, the large-scale construction of data centers has become an important development strategy for major technological nations [1]. However, the current trend of miniaturization, integration and high density of electronic devices drives up the heat flow density of data centers, resulting in the traditional air-cooled cooling mode approaching its theoretical heat transfer limit. Moreover, the electrical power consumption of the corresponding thermal management system even reaches 35% to 50% of the total data center energy consumption [2,3]. Given this, developing efficient and reliable cooling technology has become an essential guarantee for decreasing green data centers' power usage effectiveness (PUE).

Compared to traditional forced air-cooled solutions, the liquid cooling solution is known to be able to increase the cooling capacity several times [4,5]. Among them, the cold plate solution with indirect liquid-cooling has gained much attention and has some commercial examples. Parida et al. [6] explored the cooling capacity of server racks using conventional forced air-cooled and liquid-cooled plate solutions, respectively. The results show that a single-phase liquid-cooling solution can reduce cooling power consumption by more than 90% compared to typical forced air-cooling servers. However, as the coolant in cold plates is mostly electrically conductive, it has serious reliability issues such as leakage and contamination [7,8]. In contrast, the efficient cooling performance of immersion cooling solutions, where the coolant is in direct contact with power devices, not only improves the efficiency and stability of servers, but also allows for increased server density to improve computing efficiency, effectively driving the development of energy and noise reduction needs in data centers [9,10]. Currently, immersion cooling technologies are divided into single-phase immersion cooling (SPIC) [11–13] and two-phase immersion cooling (TPIC) [14–16] solutions, depending on whether the coolant undergoes a phase change. Eiland et al. [17] explored the heat transfer performance of a server submerged in mineral oil. The results im-

* Corresponding authors at: Jiangsu Key Laboratory of Micro and Nano Heat Fluid Flow Technology and Energy Application, School of Environmental Science and Engineering, Suzhou University of Science and Technology, Suzhou, Jiangsu 215009, China.

E-mail address: cbzhang2004@dingtalk.com (C. Zhang).

Nomenclature

c_p	specific heat capacity
$C_{1\varepsilon}, C_{2\varepsilon}, C_\mu$	model constants
D	inlet diameter
$\vec{g}$	gravitational acceleration vector
G_b	generation of turbulence kinetic energy due to buoyancy
G_k	generation of turbulence kinetic energy due to mean velocity gradients
h	average heat transfer coefficient
k	turbulent kinetic energy
l	hydraulic diameter
L	enthalpy
N	number of grid nodes
$\overline{Nu}$	Nusselt number
$\bar{p}$	mean pressure
P	CPU power
$P_{cooling}$	input power of the cooling system
P_{ICT}	total electrical power consumption
P_{IT}	total power of IT equipment
Q_{in}	volume flow rate
Re	Reynolds number
R_ε	correction factor of the ε equation
Ste	Stephan number
T	temperature
$\bar{T}$	mean temperature
T'	temperature pulsation
T_0	initial temperature
T_{in}	inlet temperature of cooling coolants
T_p	average temperature of cooling coolants
$\bar{u}_i$	mean velocity component
u'	velocity pulsation
V	velocity value
x_i	coordinate axis

Greek symbols

$\alpha_k, \alpha_\varepsilon$	model constants
β	model constant
Δp	flow resistance loss
ΔT_{CPU}	temperature augment of CPU
ε	turbulent kinetic energy dissipation rate
η	expansion parameter
θ	thermal transport temperature difference from the reference state
λ	thermal conductivity
μ	dynamic viscosity
μ_{eff}	effective kinetic viscosity coefficient
ξ	coefficient of thermal expansion
ρ	density
σ_T	temperature uniformity index
Σ	outer walls of immersion cooling tanks

Abbreviation

CPU	central processing unit
COP	coefficient of performance
DIMM	dual inline memory module
FOM	figure of merit
IT	information technology
MCOM	mixed convection operating mode
NCOM	natural convection operating mode
PCB	printed circuit board
PCH	platform controller hub
PSU	power supply unit

PUE	power usage effectiveness
RC	Retimer Card
SPIC	single-phase immersion cooling
TPIC	two-phase immersion cooling
VCOM	virtual convection operating mode
3D	three-dimensional

ply that the SPIC system reduces the thermal resistance by 34.4% compared to conventional air-cooling solutions and achieves a PUE as low as 1.03. On the other hand, Liu et al. [18] explored the thermal characteristics of a TPIC system under different loads numerically. The conclusions imply that the coefficient of performance (COP) for a TPIC solution could reach 26.7, and the PUE is reduced to 1.037. To compare the performance of SPIC and TPIC systems, Kanbur et al. [19] conducted an experimental study on data center serves and revealed that the COP of TPIC systems is 72.2–79.3% higher than those of the SPIC system. Although the thermal characteristics of TPIC systems are better than those of SPIC systems, it is relatively poor in terms of investment cost, safety, and maintainability [20]. Therefore, the SPIC system is more suitable for large-scale and commercial applications than the TPIC system.

A comprehensive literature survey shows that existing research on immersion cooling technology is mainly concerned with the flow and heat transfer mechanisms, the immersion cooling medium, the design of immersion cooling systems, and the immersion cooling applications. Among these, extensive theoretical and experimental research has been conducted on SPIC systems in the available literature [21–23]. Given the direct contact between power devices and coolants in a SPIC system, the thermal properties of immersion coolants directly determine the heat dissipation capacity of immersion cooling systems. Jithin et al. [24] evaluated the cooling performance of deionized water, mineral oil, and an engineered fluid in a SPIC system through numerical simulations. Their conclusions indicate that the cooling influence of a coolant is closely related to its viscosity. Compared to mineral oil and engineered fluids, the lower viscosity of deionized water resulted in a 74.4 and 69.4% reduction in the temperature rise rate of lithium batteries, respectively. Saylor et al. [25] evaluated the thermal performance of several candidate immersion coolants by defining the figure of merit (FOM) of the coolant. The results indicate that chlorinated fluorocarbon R-113 has better heat dissipation capabilities, and the increased operating temperature improves the convective heat transfer efficiency by 5–20%. The operating conditions of SPIC systems also have a significant influence on the flow and thermal transport characteristics. Cheng et al. [26] investigated the role of inlet coolant velocities in the thermal performance of SPIC systems. Their results indicate that the increase in the flow rate decreases the temperature of central processing units (CPUs) and improves heat transfer uniformity. Wang et al. [27] experimentally explored the roles of volume flow rate and immersion depth of mineral oils in the cooling capacity of SPIC systems for lithium batteries. The results mean that increasing the flow rate and immersion depth contributes to a reduction in energy consumption and results in a 9.1% reduction in the maximum battery temperature. Apart from this, SPIC solutions can be subdivided into pump-driven and buoyancy-driven SPIC systems due to the differences in drive mechanisms. Through experimental studies, Li et al. [28] surveyed the cooling capacity of a buoyancy-driven SPIC system versus a forced air-cooled solution. Their conclusions indicate that the peak temperature decreases by 23.4% compared to the forced air-cooled system. Pambudi et al. [29] designed a pump-driven SPIC system to address the issue of high electrical power consumption in data centers. Experimental results reveal that the pump-driven

SPIC system results in a 21.7% reduction in the maximum CPU temperature compared to the conventional air-cooling solution.

From a comprehensive review of the existing literature, it can be found that current research on SPIC systems is mainly focused on coolant selection [23,24], operating conditions [25,26], structural optimization [30], and flow distribution [31]. Although considerable work has been performed to investigate the influences of the immersion coolant type, volumetric flow rate, and inlet temperature on the heat dissipation capacity of data center servers, the mechanisms underlying the immersion cooling processes in SPIC systems are less well understood. Particularly, the mechanisms driving the differences in flow and heat transfer characteristics within the buoyancy-driven and pump-driven SPIC systems are not precise. In addition, further research is needed into the dynamic thermal transport properties, flow resistance characteristics, and their influence mechanisms on energy efficiency levels of SPIC systems.

Regardless of other research expansions in immersion cooling technology, the focus of this paper is on the flow and heat transfer mechanisms of SPIC systems. To further investigate the internal thermal characteristics and the intrinsic driving mechanism of SPIC systems, steady-state three-dimensional (3D) mathematical models of the flow and thermal transport processes in pump-driven and buoyancy-driven immersion cooling tanks are developed, respectively. The heat transfer characteristics, flow resistance, and energy consumption levels of immersion cooling tanks in three operating modes are investigated. Moreover, the immersion cooling mechanism of pump-driven and buoyancy-driven SPIC systems is revealed. Apart from this, an in-depth analysis of the influence mechanisms of coolant flow rate, inlet temperature, and CPU power on immersion cooling capacities is also presented, thus guiding the design and operation of SPIC systems in practical engineering.

2. Model methodology and validation

2.1. Model configuration

The SPIC system used for data centers can usually be divided into two types, i.e., pump-driven and buoyancy-driven SPIC systems. Fig. 1 plots the system architecture of two different SPIC systems. Among them, the pump-driven SPIC system mainly consists

of the immersion cooling tank, server array, internal pump, heat exchanger, external pump, and cooling tower. During normal operations, the immersion coolant enters the immersion cooling tank from the bottom and absorbs the heat generated by server motherboards, then leaves from the side wall to exchange heat with the cooling water in outer circulation loops through a heat exchanger, and finally releases thermal consumption into the surrounding environment through a cooling tower. Fig. 1(b) shows the SPIC system driven by buoyancy force, which mainly includes the immersion cooling tank, server array, cooling coils, pump, and cooling tower. Unlike the pump-driven one, the buoyancy-driven SPIC system mainly relies on the natural convection caused by the thermal expansion of immersion coolants to bring waste heat from servers to up parts of the immersion cooling tank, then transport the heat to the external circulation loops through heat exchange with cooling coils, and finally release thermal consumption to the surrounding environment in a cooling tower as well. Obviously, the thermal performance of SPIC systems with two different drive mechanisms mainly depends on the heat dissipation capability of immersion cooling tanks.

Given this, this work conducts a comparative investigation on the cooling capacity of two different immersion cooling tanks mentioned above. To save consumption time and computational cost, a simplified immersion cooling tank containing only one server motherboard is used here as the computational domain. Fig. 2 shows the geometry of the pump-driven and buoyancy-driven immersion cooling tanks. The overall dimensions of an immersion cooling tank are 550 mm to 680 mm in length, 55 mm in width, and 1040 mm in height. Besides, the buoyancy-driven immersion cooling tank also includes a cooling coil with an inner diameter of 12 mm and a total length of 9200 mm. It should be noted that the immersion coolant herein uses the fluorinert electronic liquid produced by 3 M as the cooling coolant, while the coolant in the cooling coil is water, and their thermal properties are shown in Table 1. Moreover, the server motherboard is commercially available in the current market, mainly composed of the printed circuit board (PCB), CPU, dual inline memory module (DIMM), power supply unit (PSU), platform controller hub (PCH), and Retimer Card (RC), as shown in Fig. 2(c). Table 2 gives the specific size and power consumption of each electronic device. Notably, the wall thickness of immersion cooling tanks and the ambient heat loss are neglected in this paper to simplify the calculation.

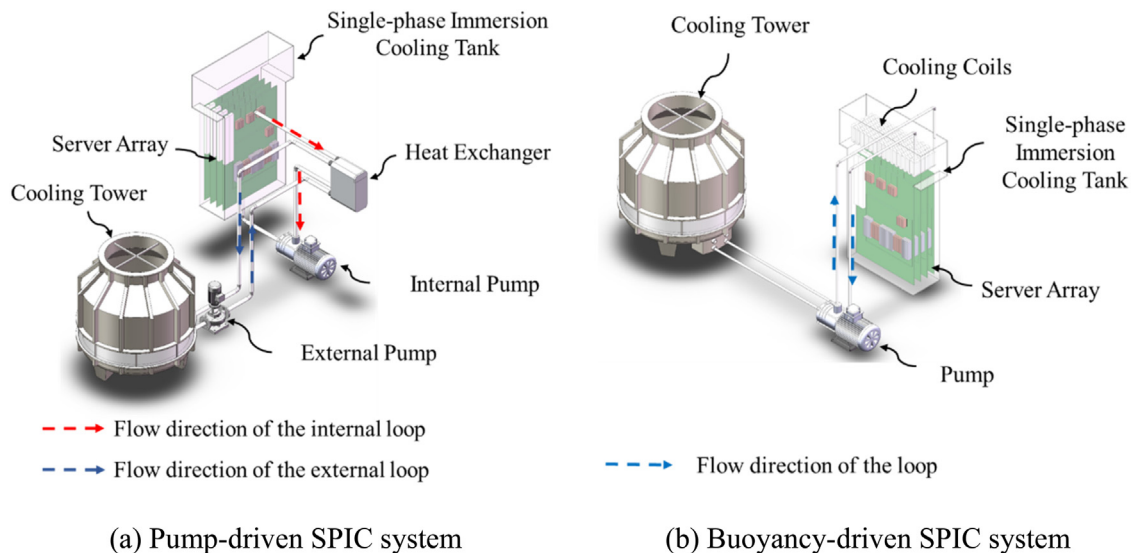
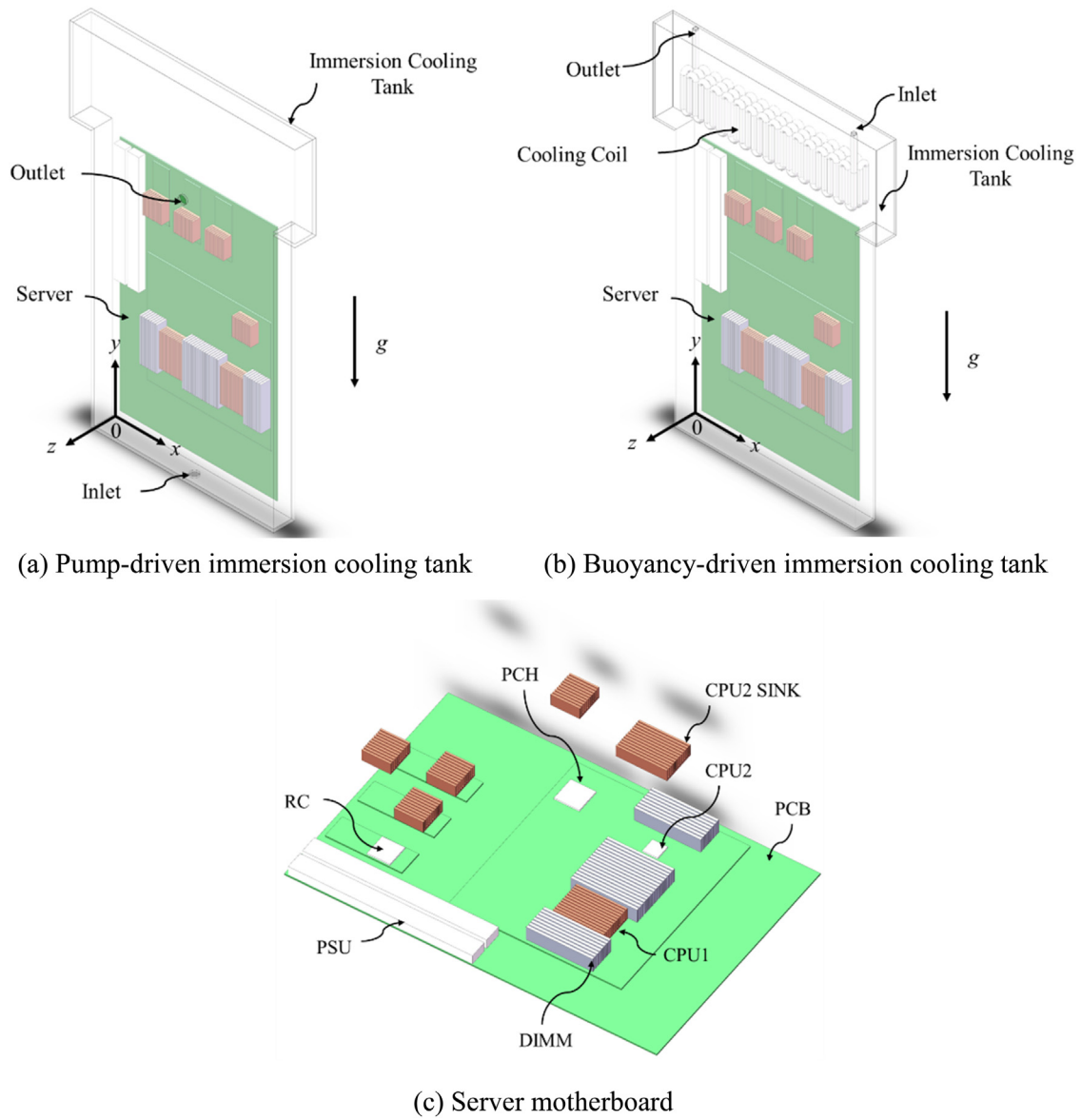


Fig. 1. Principle diagram of different SPIC systems for data centers.
(a) Pump-driven SPIC system (b) Buoyancy-driven SPIC system.

**Fig. 2.** Geometry of different immersion cooling tanks.

(a) Pump-driven immersion cooling tank (b) Buoyancy-driven immersion cooling tank (c) Server motherboard.

2.2. Governing equations

In this work, a 3D steady-state mathematical model of flow and heat transfer processes is constructed to investigate the cooling capacity of pump-driven and buoyancy-driven immersion cooling tanks. To simplify numerical calculations, the below assumptions are given here: 1) the flow process is steady-state, constant, and incompressible; 2) the effect of thermal radiation is ignored; 3) the flow state is turbulent; 4) the viscous dissipative heat is ignored;

5) the Boussinesq hypothesis is used to relate the Reynolds stresses to the mean velocity gradients.

Based on the above hypotheses and three conservation equations, the Reynolds averaged Navier-Stokes model of flow and thermal transport processes for fluorinert electronic liquids inside pump-driven immersion cooling tanks and cooling water in the cooling coil of buoyancy-driven immersion cooling tanks can be described by

Table 1
Thermal properties of coolants in this paper.

Thermophysical properties	Fluorinert electronic liquid	Water
Boiling point (1atm) (°C)	128	100
Density (kg/m ³)	$\rho=1878-2.455T$	998.2
Dynamic Viscosity $\times 10^3$ (Pa·s)	1.4	1.004
Specific thermal capacity [J/(kg·°C)]	1100	4183
Thermal conductivity [W/(m·°C)]	0.066	0.599
Thermal expansion coefficient (1/°C)	0.0014	0.0018

Table 2
Size and power of electronic devices on the server motherboard.

Device Name	Geometric dimensions (mm)	Power (W)	Quantity
CPU	30 × 30 × 3	270	2
DIMM	130 × 5 × 30	15.1	28
PCH	50 × 50 × 3	30.0	1
PSU	365 × 35 × 20	99.2	2
RC	45 × 45 × 3	46.9	3

Continuity equation:

$$\frac{\partial(\rho \bar{u}_i)}{\partial x_i} = 0 \quad (1)$$

Momentum equation:

$$\frac{\partial}{\partial x_j}(\rho \bar{u}_i \bar{u}_j) = -\frac{\partial \bar{p}}{\partial x_i} + \mu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} + \rho \bar{g} + \frac{\partial}{\partial x_j}(-\rho \overline{u'_i u'_j}) \quad (2)$$

Energy equation:

$$\frac{\partial(c_p \rho \bar{u}_i \bar{T})}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\lambda \frac{\partial \bar{T}}{\partial x_i} - \rho c_p \overline{u'_i T'} \right) \quad (3)$$

where ρ denotes density, $\bar{u}_i$ denotes mean velocity component in each axial x_i , $\bar{T}$ represents mean temperature, $\bar{p}$ denotes mean pressure, μ denotes dynamic viscosity, $\bar{g}$ represents gravitational acceleration vector, λ represents thermal conductivity, c_p denotes specific heat capacity, u'_i , u'_j are the velocity pulsation in different directions, T' is the temperature pulsation. As the simplest complete turbulence model, the two-equation model is widely used in practical computational fluid dynamics, including the *standard k- ε* model, *RNG k- ε* model, and *Realizable k- ε* model. Among these, the *RNG k- ε* model considers the effect of eddies on turbulence and provides an analytical formula for the turbulent Prandtl number, thus improving the accuracy of the turbulence calculation. Furthermore, the *RNG* theory provides an analytical formulation that takes into account the viscosity of low Reynolds number flows. These features allow the *RNG k- ε* model to have higher confidence and accuracy over a broader range of flows than the *standard k- ε* model. On the other hand, the *Realizable k- ε* model is essentially the same as the *RNG k- ε* model, but its computational cost and time consumption are relatively large. Based on the trade-off between computational accuracy and computational cost, the *RNG k- ε* model [32,33] is used to describe the turbulent flow state in this paper.

k equation:

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho k \bar{u}_i)}{\partial x_i} = \frac{\partial}{\partial x_j} \left(\alpha_k \mu_{\text{eff}} \frac{\partial k}{\partial x_j} \right) + G_k + G_b - \rho \varepsilon \quad (4)$$

ε equation:

$$\begin{aligned} \frac{\partial(\rho \varepsilon)}{\partial t} + \frac{\partial(\rho \varepsilon \bar{u}_i)}{\partial x_i} &= \frac{\partial}{\partial x_j} \left(\alpha_\varepsilon \mu_{\text{eff}} \frac{\partial \varepsilon}{\partial x_j} \right) \\ &+ C_{1\varepsilon} \frac{\varepsilon}{k} (G_k + C_{3\varepsilon} G_b) - C_{2\varepsilon} \rho \frac{\varepsilon^2}{k} - R_\varepsilon \end{aligned} \quad (5)$$

where k denotes turbulent kinetic energy, ε represents turbulent kinetic energy dissipation rate, G_k and G_b represent the generation of turbulence kinetic energy due to the mean velocity gradients and buoyancy, respectively, μ_{eff} is the effective kinetic viscosity coefficient, R_ε is the correction factor of ε equation, which can be calculated by

$$R_\varepsilon = \frac{C_\mu \rho \eta^3 (1 - \eta/\eta_0) \varepsilon^3}{1 + \beta \eta^3} \frac{\varepsilon^3}{k} \quad (6)$$

where η is the expansion parameter (i.e., the ratio of the turbulent to the mean-strain time scales), and other model constants are given in Table 3.

To simplify calculations, each electronic component in this paper is equated to a bulk heat source with a uniform heat flow

density in the numerical calculations. For the server motherboard, only heat conduction is considered and its heat transfer equation is written as

$$\lambda \frac{\partial}{\partial x_i} \left(\frac{\partial T}{\partial x_i} \right) = 0 \quad (7)$$

To describe the buoyancy-driven immersion cooling, the Boussinesq assumption is used herein since the density of the immersion coolant in this paper does not vary significantly with temperature. Given this, the governing equations of flow and thermal transport processes of internal fluorinated electronic liquids are given by

Continuity equation:

$$\frac{\partial(\rho u_i)}{\partial x_i} = 0 \quad (8)$$

Momentum equation:

$$\frac{\partial}{\partial x_j}(\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right] + \rho \bar{g} \theta \xi \quad (9)$$

Energy equation:

$$\frac{\partial(\rho u_i T)}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\frac{\lambda}{c_p} \cdot \frac{\partial T}{\partial x_i} \right) \quad (10)$$

where ξ denotes the coefficient of thermal expansion, and θ represents the thermal transport temperature difference from the reference state.

The initial condition for the overall immersion cooling tank is given by

$$T_{t=0} = T_0 \quad (11)$$

The boundary condition of the outer walls (i.e., Σ) for immersion cooling tanks is described by

$$-\lambda \left(\frac{\partial T}{\partial \vec{n}} \right)_\Sigma = 0 \quad (12)$$

The inlet conditions for immersion cooling tanks and the cooling coil are given by

$$Q_{\text{in}} = \text{constant}; T_{\text{in}} = \text{constant} \quad (13)$$

where T_0 is the initial temperature of the immersion cooling tank, Q_{in} and T_{in} are the volume flow rate and inlet temperature of cooling coolants (i.e., water and fluorinated electronic liquid). In the current work, the initial temperature of immersion cooling tanks is $T_0 = 20^\circ\text{C}$. Moreover, the liquid-solid interface adopts no-slip boundary and conjugate heat transfer boundary conditions. To explore the influences of operation conditions, the variation ranges of inlet temperature and volume flow rate herein are $20\text{--}40^\circ\text{C}$ and $3.7\text{--}67.82\text{ L/min}$, respectively. Besides, the default value of inlet temperature and volume flow rate is 20°C and 13.56 L/min , respectively.

2.3. Numerical approach and verification

To investigate the SPIC system for data centers, the ICEM CFD package of ANSYS 2021 is used to mesh the computational domain of different immersion cooling tanks. In view of the complex geometric model of server motherboards, an unstructured mesh is chosen in this paper, and the computational grid for key electronic devices (e.g., CPU, etc.) is locally refined. In this context, the finite volume algorithm is applied to numerically simulate the flow and thermal transport characteristics of immersion cooling tanks by the commercial software FLUENT 2021. Considering the complexity of the flow field in immersion cooling tanks, the *RNG k- ε* model is chosen here for simulations of the turbulent flow process. To deal with the turbulent flow problems involved in this paper, the finite volume method is used for the numerical solution.

Table 3
Constants related to the *RNG k- ε* model.

α_k	α_ε	$C_{1\varepsilon}$	$C_{2\varepsilon}$	C_μ	η_0	β
1.393	1.393	1.42	1.68	0.0845	4.38	0.012

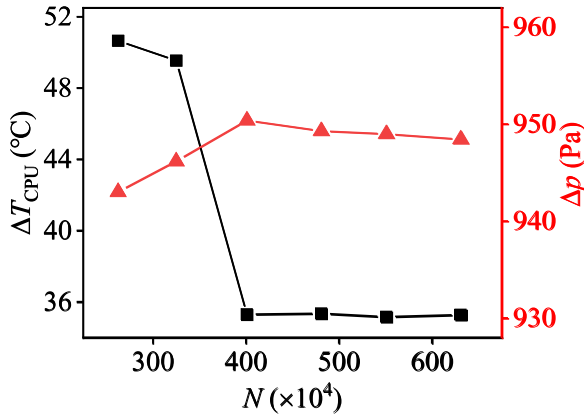


Fig. 3. Grid-independence verification.

The pressure-velocity coupling in numerical solution processes is realized by the pressure correction method based on the SIMPLE algorithm. In the spatial discretization of computational domains, the pressure term is discretized using the PRESTO algorithm, the momentum, energy, turbulent kinetic energy, and energy dissipation rate equations are discretized in a second-order windward format, and the corresponding relaxation factors are given as 0.3 for pressure, 1 for density, 0.7 for momentum, and 0.9 for energy. To improve the calculation accuracy, the pressure-based double precision solver is chosen in this paper. Finally, when the residuals of the energy equation are less than 10^{-6} and the residuals of other equations are less than 10^{-4} , the iterative calculation is stopped and the simulation is considered to have converged.

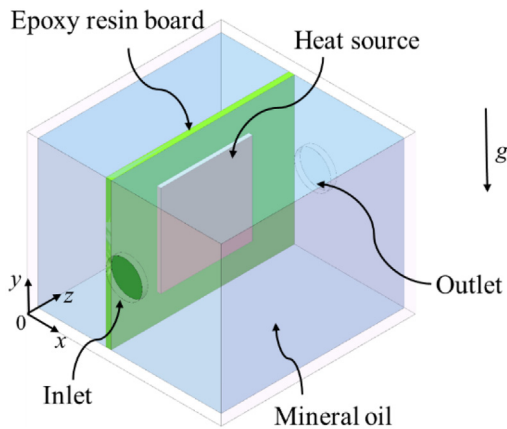
To ensure the reliability and reproducibility of numerical results, the grid independence is verified by taking the pump-driven immersion cooling tank as an example. For this purpose, the flow and thermal transport capacity of a pump-driven immersion cooling tank are calculated under different grid systems. The temperature augment of CPU1 (i.e., ΔT_{CPU}) and the flow resistance loss (i.e., Δp) for the immersion cooling tank under different grid systems are given in Fig. 3. As shown in the figure, the number of grid nodes has almost no effect on the temperature augment of CPU1 and flow resistance loss when the number of grid nodes exceeds 4,812,822. Therefore, given the weighing between simulation accuracy and computational resource consumption, the number of grid nodes for all numerical calculations in this paper should be at least 4,812,822.

2.4. Model validation

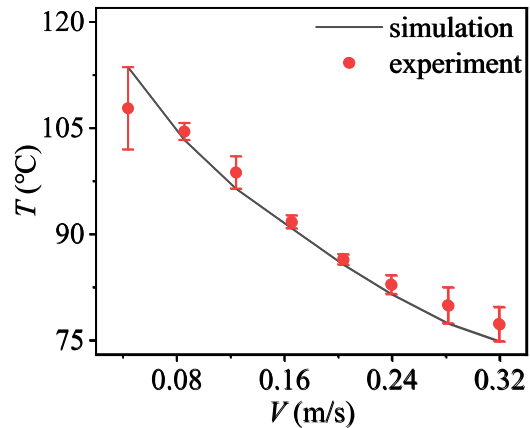
The numerical model validation is essential to support the reliability and credibility of conclusions. Therefore, a comparative study of the immersion cooling experiment by Luo et al. [34] is presented herein. In their experiment, an epoxy resin plate with a heat source is applied to simulate the actual server motherboard, and the immersion coolant is mineral oil. Moreover, the dimensions of the heat source are 50 mm $\times$ 50 mm $\times$ 2 mm, and the maximum power consumption is 60 W. The experiments of Luo et al. [34] are replicated herein by numerical simulations under the same experimental geometries and working conditions. Fig. 4(a) shows the physical model constructed in this paper. Moreover, Fig. 4(b) gives a comparison between simulation results and experimental data in the average temperature of the heat source. In this comparison, the heat source power is constant and the inlet temperature of mineral oil is 25 $^\circ\text{C}$. As plotted in the figure, the predicted heat source average temperature agrees well with the experimental results of Luo et al. [34]. According to the data analysis, the deviation between experimental and numerical results at different flow velocities is only 5.44%. This is mainly due to the unavoidable measurement errors in the experimental process and the simplification of heat sources to a bulk heat source with uniform heat flow density in the simulation. In conclusion, although there is a small deviation in the simulation results compared to experiments, the error range is within the permissible range of scientific research. Therefore, the current model can be considered reliable and accurate, and subsequent related studies can be conducted based on the present model.

3. Results and discussion

To explore the performance of different SPIC systems, the temperature response, flow characteristics, and energy efficiency levels inside pump-driven and buoyancy-driven immersion cooling tanks are investigated deeply in this paper. Notably, the operating modes in pump-driven immersion cooling tanks herein are subdivided into forced convection with and without consideration of buoyancy forces, i.e., the mixed convection operating mode (MCOM) with nature convection and virtual convection operating mode (VCOM) without nature convection. In addition, the buoyancy-driven immersion cooling tank corresponds to the natural convection operating mode (NCOM).



(a) Physical model



(b) Comparison of experiment and simulation

Fig. 4. Model validation for an immersion cooling experiment. (a) Physical model (b) Comparison of experiment and simulation.

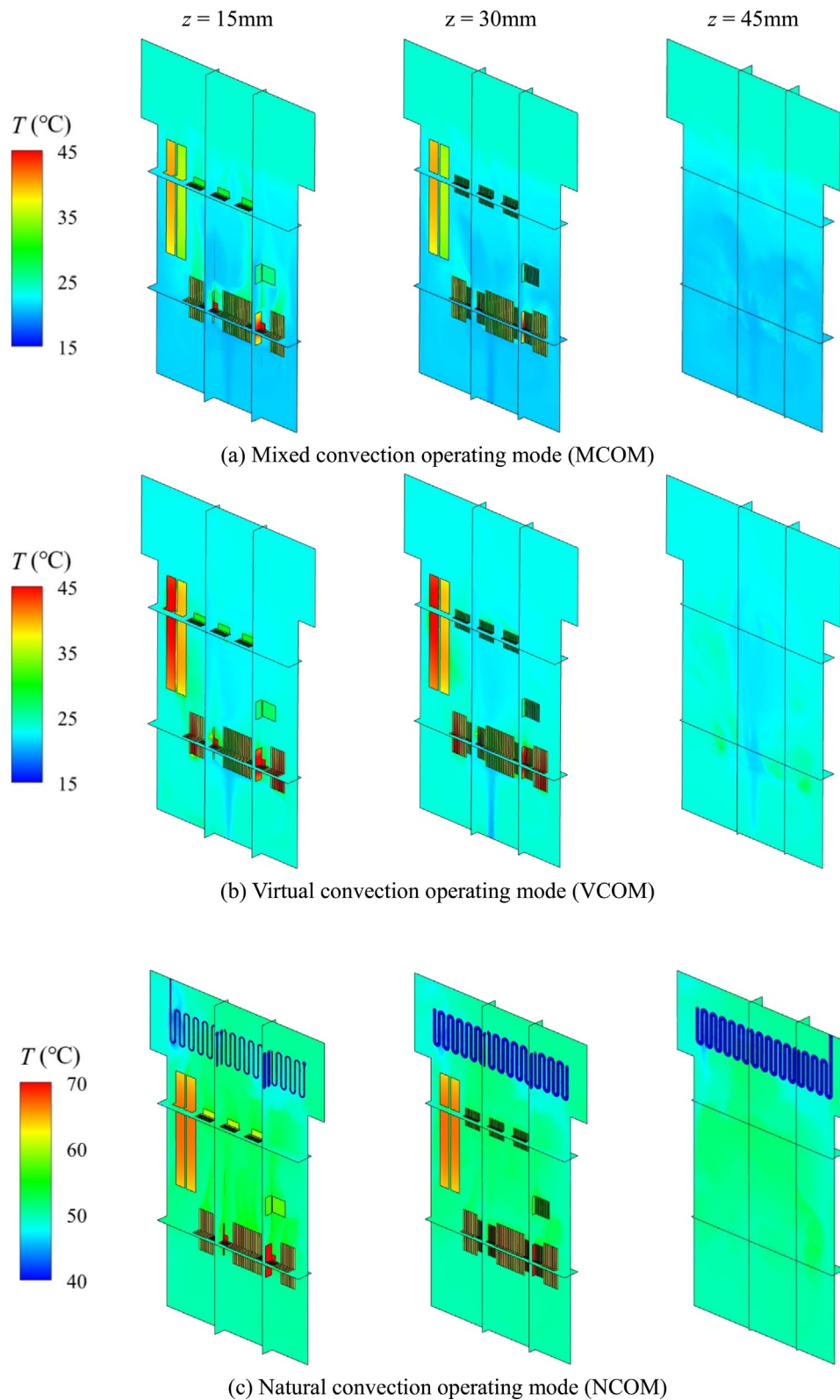


Fig. 5. Temperature distribution inside immersion cooling tanks in different operating modes (a) Mixed convection operating mode (MCOM) (b) Virtual convection operating mode (VCOM) (c) Natural convection operating mode (NCOM).

To quantitatively reveal the influence mechanism of immersion cooling tanks, this paper also introduces two dimensionless numbers, i.e., Stephan number Ste and Reynolds number Re .

The Stephan number, Ste , is defined as the ratio of sensible heat to latent heat. Notably, there is no phase change process in this

paper. Therefore, it is possible to assess the magnitude of the sensible heat by Ste because the latent heat of an immersion coolant is constant. Given this, Ste can be calculated as

$$Ste = c_p(T_p - T_{in})/L \quad (14)$$

The Reynolds number, Re , can be applied to describe the inlet flow state of immersion cooling tanks, which can be expressed as

$$Re = \rho V D / \mu \quad (15)$$

where T_p denotes the average temperature of the cooling coolant in immersion cooling tanks, L and V denote the enthalpy and inlet velocity value of cooling coolants, respectively, and D is the inlet diameter of immersion cooling tanks.

3.1. Performance comparison of SPIC systems with three operating modes

To provide a fair comparison, the following analysis in this paper is based on the premise that two different cooling coolants have the same inlet conditions of $Re = 16,000$ and $Ste = 0.32$. In addition, two chip sections along the x -direction ($x = 225, 410$ mm) and y -direction ($y = 285, 655$ mm), respectively, and three sections in the z -direction ($z = 15, 30, 45$ mm) are selected as reference states for the analysis in this paper.

3.1.1. Heat transfer performance

Fig. 5 gives the temperature distribution inside immersion cooling tanks under different operating modes. As plotted in the figure, the temperature gradient along the CPU cross-section ($z = 15$ mm) is higher than that of other electronic devices ($z = 30$ mm) and the fluorinert electronic liquids ($z = 45$ mm) cross-section. Compared with the VCOM case, the average temperature of fluorinert electronic liquids in the MCOM case reduces by 3.5%, and the maximum temperature of CPU1 decreases by 8.1% as well. Moreover, the temperature of fluorinert electronic liquids increases slowly along the anti-gravity direction in the MCOM case. This phenomenon arises due to the generation and development of natural convection, which enhances the perturbation of fluorinert electronic liquids and improves thermal transport performance. Simultaneously, the fluorinert electronic liquid is subject to buoyancy forces and temperature stratification occurs inside the immersion cooling tank in the MCOM case. On the other hand, compared with the MCOM and VCOM cases, the average temperature of fluorinert electronic liquids in the NCOM case increases by 124.94 and 117.1%, respectively, and the maximum CPU1 temperature increases by 66.6% and 52.8%, respectively. This phenomenon indicates the poor performance of immersion cooling tanks in cases of NCOM. From the above analysis it is clear that, for the same operating conditions and inlet boundaries, the heat dissipation of immersion cooling tanks in the NCOM case is less effective than in the MCOM and VCOM cases. More importantly, the thermal performance of immersion cooling tanks in the MCOM case is better than in the VCOM case, which suggests that the influence of natural convection on the thermal transport characteristics in pump-driven immersion cooling tanks cannot be ignored.

Fig. 6 further presents the temperature uniformity index of immersion cooling tanks in different operating modes to quantify the impact of different thermal transport mechanisms on thermal uniformity. Notably, the temperature uniformity index in this paper is based on the standard deviation (i.e., σ_T) of the temperature at 24 different locations ($x = 137.5, 275, 412.5$ mm; $y = 200, 400, 600, 800$ mm; $z = 20, 40$ mm) inside the immersion cooling tank. It should be noted that the locations of these temperature monitoring points have been selected to ensure that the average temperature of all temperature monitoring points is close to the average temperature of fluorinert electronic liquids inside the immersion cooling tanks. In this context, the larger the temperature uniformity index, the worse the thermal transport uniformity for the immersion cooling tanks. It is evident that the temperature uniformity indices of immersion cooling tanks in the MCOM and NCOM cases are similar and both higher than those in the VCOM case.

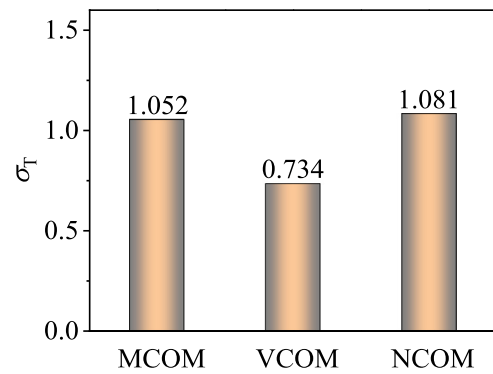


Fig. 6. Temperature uniformity index of immersion cooling tanks in different operating modes.

According to data analysis, the temperature uniformity index of immersion cooling tanks in the VCOM case is 30.2% lower than in the MCOM case. This interesting phenomenon indicates that the heat transfer uniformity inside pump-driven immersion cooling tanks decreases instead because of the natural convection influence. Thus, although the natural convection excited by the thermal expansion of fluorinert electronic liquids strengthens the heat transfer characteristics to some extent, the consequent temperature stratification phenomenon results in a deterioration in the temperature uniformity of immersion cooling tanks. A close look at Fig. 6 also shows that the temperature uniformity index of immersion cooling tanks in the MCOM case differs from that of the NCOM case by only 2.7%, demonstrating the significant influence of natural convection on the uniform heat transfer in the immersion cooling tank.

3.1.2. Dynamic flow characteristics

To investigate the flow characteristics of different immersion cooling tanks, the velocity field characteristics in different operating modes are given in Fig. 7. Obviously, the fluorinert electronic liquids' cross-section ($z = 45$ mm) of immersion cooling tanks in different operating modes both have a smaller velocity gradient than other cross-sections ($z = 30, 15$ mm) due to the absence of blocking effect by server motherboards, and there is no obvious mutation in the velocity magnitude distribution as well. Compared with the pump-driven immersion cooling tanks in the MCOM and VCOM cases, the velocity of fluorinert electronic liquids for the immersion cooling tank in the NCOM case is relatively small, but its velocity distribution is relatively uniform and the flow disturbances range is larger. Similarly, for pump-driven immersion cooling tanks, the convection intensity and disturbance range in the MCOM case that considers the natural convection effect are significantly higher than those in the VCOM case, accounting for the improved thermal performance of electronic devices. Moreover, the location of the inlet and outlet of immersion cooling tanks and server motherboard layout also has an important impact on the internal flow field characteristics. Further observation reveals that the uneven distribution of the inlet flow in pump-driven immersion cooling tanks results in a much higher velocity above the inlet than on both sides of immersion cooling tanks. In addition, due to the asymmetric arrangement of electronic devices on server motherboards, a bypass channel with lower flow resistance is formed on the right side of immersion cooling tanks. In this context, the velocity on the right side of immersion cooling tanks in the NCOM case is significantly higher than the other parts. In summary, the variations in the operating mechanisms of immersion cooling tanks result in their flow characteristics being influenced differently by the device layout on the server motherboard.

For the purpose of quantitatively evaluating the influence of different thermal transport mechanisms on the immersion cooling

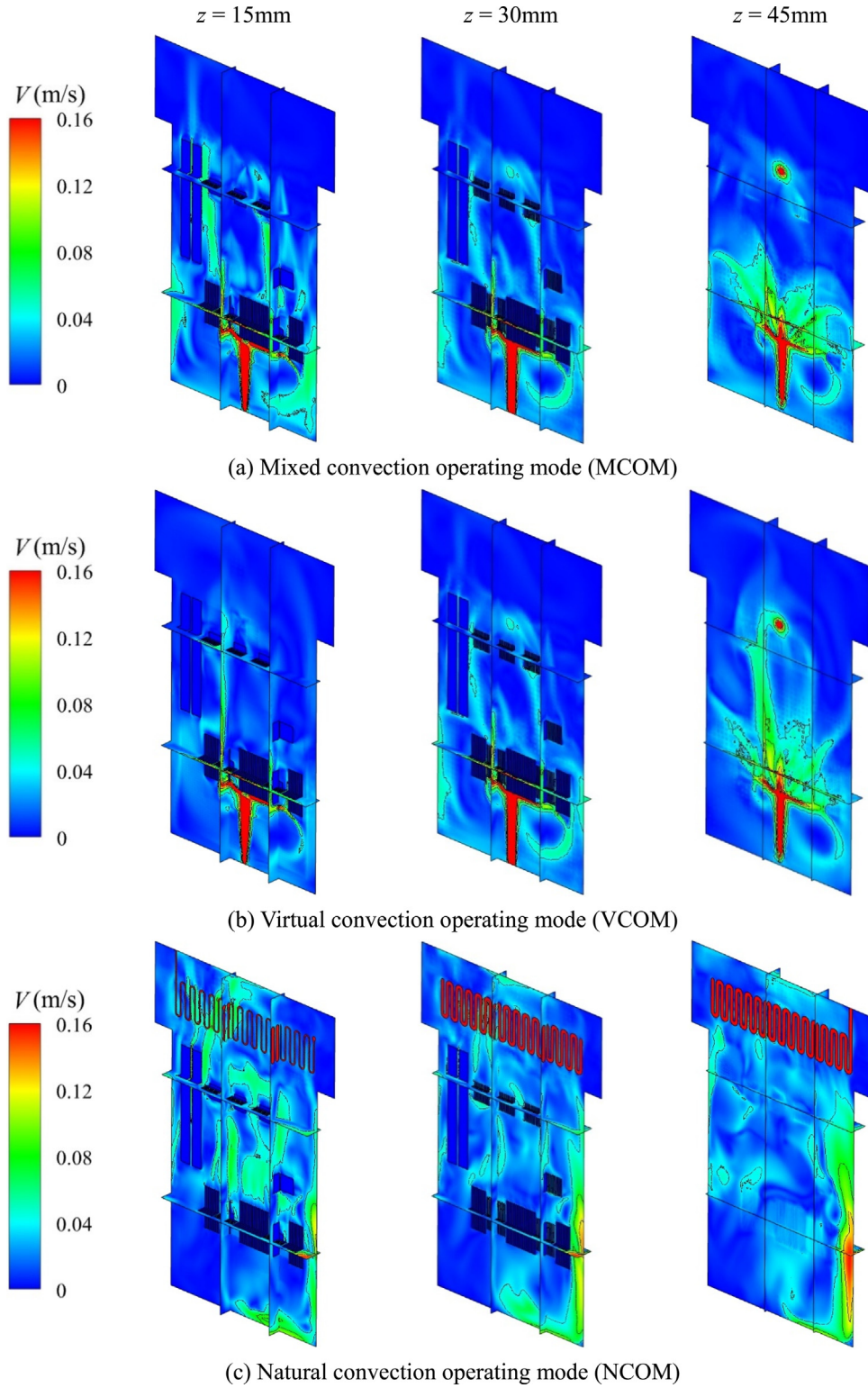


Fig. 7. Velocity distribution inside immersion cooling tanks in different operating modes

(a) Mixed convection operating mode (MCOM) (b) Virtual convection operating mode (VCOM) (c) Natural convection operating mode (NCOM).

capacity, the average Nusselt number (i.e., $\overline{Nu}$) inside immersion cooling tanks in different operating modes is selected as an evaluation index, which is defined as follows

$$\overline{Nu} = hl/\lambda \quad (16)$$

where h denotes the average heat transfer coefficient, and l represents the hydraulic diameter of immersion cooling tanks. The av-

erage Nusselt number for immersion cooling tanks in different operating modes is given in Fig. 8. As plotted in the figure, the average Nusselt number in an MCOM case is the largest and thus presents the best flow and heat transfer performance. Compared to the NCOM case, the average Nusselt number in a VCOM case is improved by 100.99%, indicating that the performance of pump-driven immersion cooling tanks is greatly improved compared with

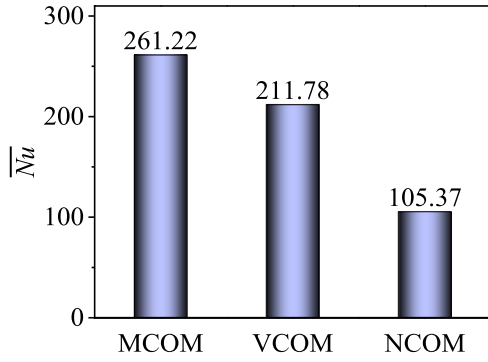


Fig. 8. Average Nusselt number of immersion cooling tanks in different operating modes.

the buoyancy-driven immersion cooling tank. In addition, the average Nusselt number inside immersion cooling tanks in the MCOM case has improved by 23.34% compared to the VCOM case. This shows that natural convection also plays a significant role in the flow and heat transfer process of pump-driven immersion cooling tanks. In this context, the proper enhancement of natural convection can improve the immersion cooling performance of pump-driven immersion cooling tanks.

3.1.3. Energy efficiency levels

To assess the energy efficiency level of SPIC systems under different operating modes, this paper selects the PUE as the evaluation index for the case when the maximum temperature (i.e., CPU1) inside different immersion cooling tanks reaches 90 °C. Notably, the PUE is calculated as the ratio of power consumption of information technology (IT) equipment to total power consumption, which can be calculated by

$$PUE = \frac{P_{ICT}}{P_{IT}} = \frac{P_{IT} + P_{cooling}}{P_{IT}} \quad (17)$$

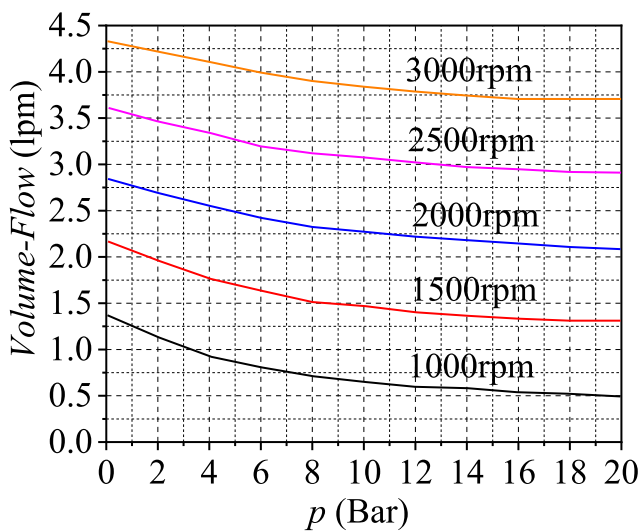
where P_{ICT} denotes the total electrical power consumption of a data center, P_{IT} denotes the total power of IT equipment in a data center, and $P_{cooling}$ denotes the input power of the cooling system. In this paper, P_{IT} denotes the total power consumption of

a server motherboard that equals 1331.9 W, while $P_{cooling}$ is the pump power of internal and external circulation pumps. It should be noted that the PUE in this paper is obtained with the same heat transfer capacity of CPUs. Thus, the magnitude of the PUE reflects the pressure drop level of immersion cooling tanks. This is because the pressure drop of immersion cooling tanks is positively correlated with the additional power consumption.

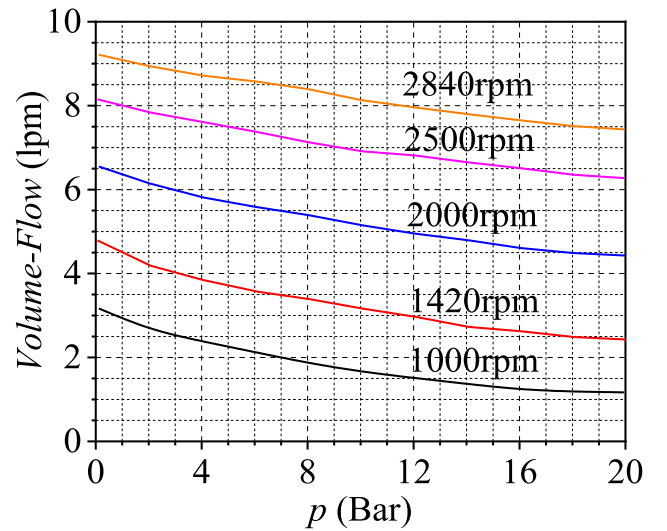
Due to the pump power mainly depending on the flow rate of immersion cooling systems as well as the flow resistance loss, this paper estimates the required pump power for immersion cooling tanks in different drive mechanisms based on the numerically simulated operating conditions and the specification parameters of the circulating pumps. For the pump-driven SPIC system, the SUOFO NP170–100 W (Rotational speed: 100~4000 rpm, Flow rate: 0~5.4 L/min) is selected for the internal and external pumps, while the SUOFO NP350–300 W (Rotational speed: 100~4000 rpm, Flow rate: 0~10.5 L/min) is selected for the buoyancy-driven SPIC system. The specification parameters of these two circulation pumps are given in Fig. 9. In addition, the heat exchanger in this paper chooses the brazed heat exchanger (Brand: CB60–50H) produced by Alfa Laval, and its heat transfer efficiency is about 98.5%. According to the above data analysis, when the maximum temperature of CPU1 reaches 90 °C, the volume flow rate of the internal circulation and external loops in the pump-driven SPIC system is about 3.7 L/min and 1.56 L/min, respectively, while the flow rate of the buoyancy-driven SPIC system reaches 8.4 L/min. In this context, the power consumption of pumps in the two SPIC systems can be obtained based on the linear analogy principle, and then the corresponding PUE is achieved as shown in Fig. 10. It is evident that the pump-driven SPIC system has the lowest PUE (i.e., 1.07), and its PUE is 11.6% lower than that of the buoyancy-driven SPIC system. The above statistical analysis fully illustrates that the pump-driven SPIC system has a higher energy efficiency than the buoyancy-driven one.

3.2. Influence factors for performance of pump-driven SPIC system

The above study shows that the heat dissipation capability and energy efficiency level of pump-driven immersion cooling tanks is better than that of the buoyancy-driven one. In order to guide



(a) SUOFO NP170-100W



(b) SUOFO NP350-300W

Fig. 9. Specification parameters of different circulation pumps
(a) SUOFO NP170–100 W (b) SUOFO NP350–300 W.

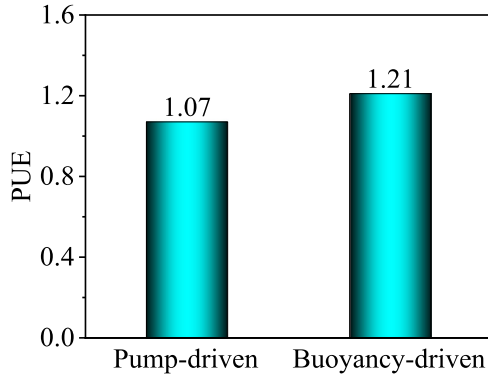
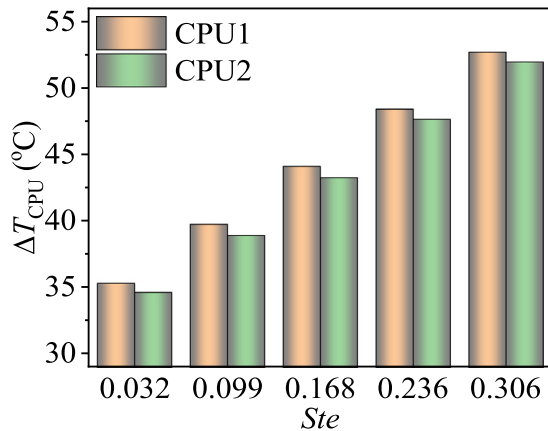


Fig. 10. Energy efficiency levels of pump-driven and buoyancy-driven SPIC systems.

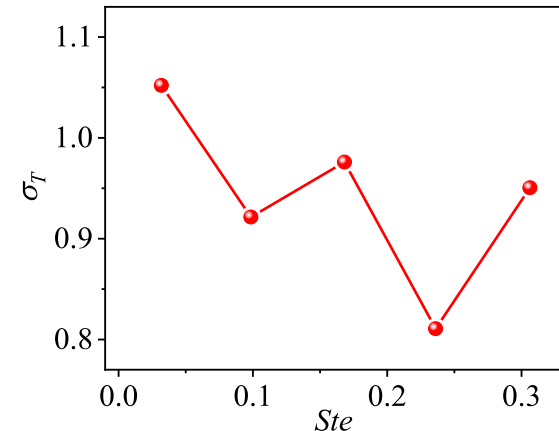
practical engineering applications, this paper further analyzes the roles of the inlet temperature, volume flow rate, and CPU power in flow and thermal transport characteristics in pump-driven immersion cooling tanks.

3.2.1. Effect of inlet temperature

To further investigate the impact of the inlet temperature of fluorinert electronic liquids on flow and thermal transport processes, Fig. 11 shows the variation of CPU temperature augment and overall temperature uniformity in pump-driven immersion cooling tanks under different Ste . As observed in Fig. 11(a), the maximum temperature augment of CPUs (i.e., CPU1 and CPU2) is positively correlated with Ste , and the maximum temperature augment of CPUs in pump-driven immersion cooling tanks increases accordingly as Ste increases. This phenomenon is mainly because the thermal exchange temperature difference between power devices and fluorinert electronic liquids decreases as Ste increases, which weakens the potential of convective heat transfer, thus making the maximum temperature augment of CPUs continuously increases. Besides, it is noticed from Fig. 11(b) that the temperature uniformity index shows an overall decreasing trend with the increase of Ste . This interesting phenomenon is because the natural convection intensity excited by the temperature variance between the upper and lower fluorinert electronic liquids weakens as the inlet temperature increases, resulting in the gradual elimination of the temperature stratification, thus enhancing the overall temperature uniformity of pump-driven immersion cooling tanks.



(a) CPU temperature augment



(b) Temperature uniformity index

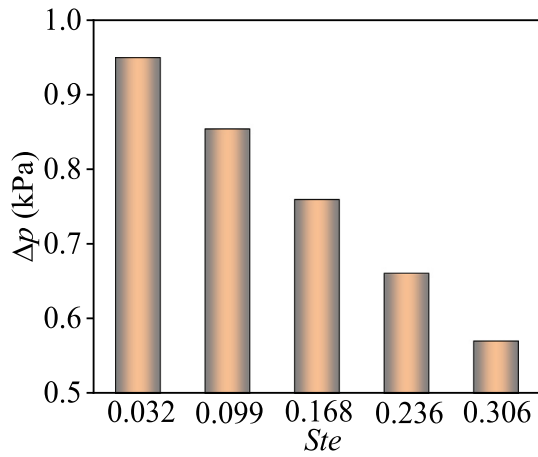
Fig. 11. Effect of inlet temperature on the temperature characteristics ($Re = 16,000$, $P = 270$ W)
(a) CPU temperature augment (b) Temperature uniformity index.

The variations of the flow resistance loss (i.e., Δp) and average Nusselt number (i.e., $\overline{Nu}$) with Ste for the pump-driven immersion cooling tank are given in Fig. 12. As plotted in Fig. 12(a), the pressure drop between the inlet and outlet shows a monotonically decreasing trend with the increase of Ste . The data analysis shows that as Ste increases from 0.032 to 0.306, the flow resistance loss of the pump-driven immersion cooling tank decreases by 39.9%. As mentioned above, this is because the increase of Ste weakens the natural convection and reduces the convective disturbance in the fluorinert electronic liquids, which in turn leads to the reduction of the pressure drop of pump-driven immersion cooling tanks. Fig. 12(b) indicates that the increase of Ste is beneficial to an increase in the average Nusselt number. As the inlet temperature changes from 20 to 40 °C, the average Nusselt number improves by 11.6%. It is noteworthy that as the Ste increases, although the corresponding thermal transport enhancement gain decreases because of the natural convection attenuation, the equivalent heat transfer coefficient of forced convection is improved and the thermal transport enhancement gain is greater, thus the average Nusselt number of pump-driven immersion cooling tanks gradually increases.

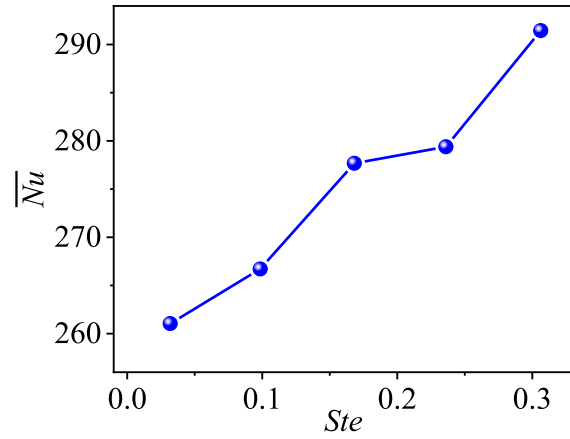
3.2.2. Effect of flow rate

To investigate the effect of the inlet flow rate of fluorinert electronic liquids on flow and thermal transport processes, Fig. 13 gives the variations of CPU temperature augment and overall temperature uniformity with Re in pump-driven immersion cooling tanks. It is obvious from Fig. 13(a) that as the inlet Re increases, the maximum temperature augment of CPUs shows a decreasing trend. Unlike the effect of inlet temperature, the difference between the maximum temperature augment of CPU1 and CPU2 gradually increases as Re rises, eventually stabilizing at around 2.3 °C. This phenomenon indicates that the role of forced convection in the thermal transport characteristics of pump-driven immersion cooling tanks is greater than that of natural convection. Moreover, the flow of fluorinert electronic liquids eventually develops into a fully developed turbulent flow as the increase of Re . This conjecture is verified in Fig. 13(b), where the enhanced convective disturbance generated with increasing Re improves the heat transfer uniformity in pump-driven immersion cooling tanks. However, as the Re increases further, there is little change in its temperature response characteristics due to the full development of turbulence.

Fig. 14 depicts the trend of the flow and thermal characteristics of pump-driven immersion cooling tanks with the inlet Re . As observed from Fig. 14(a), the variation of the flow resistance loss with



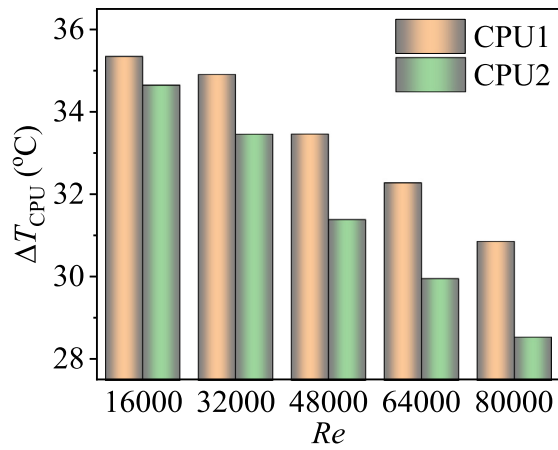
(a) Flow resistance loss



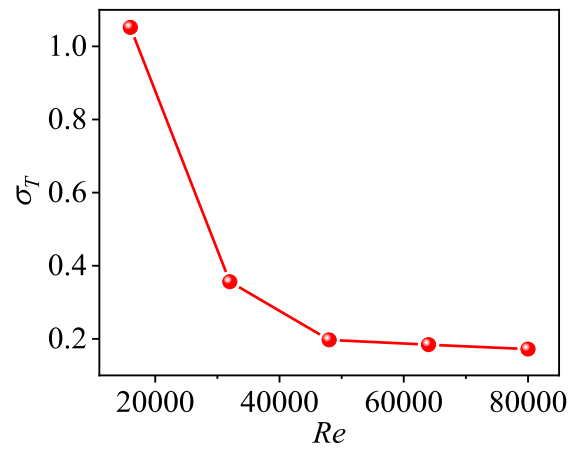
(b) Average Nusselt number

Fig. 12. Effect of inlet temperature on the flow and thermal characteristics ($Re = 16,000$, $P = 270$ W)

(a) Flow resistance loss (b) Average Nusselt number.



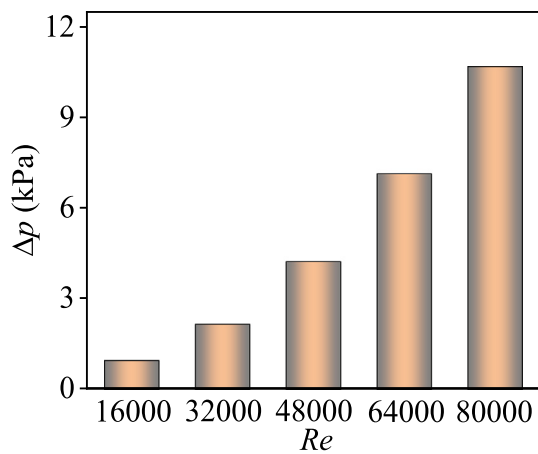
(a) CPU temperature augment



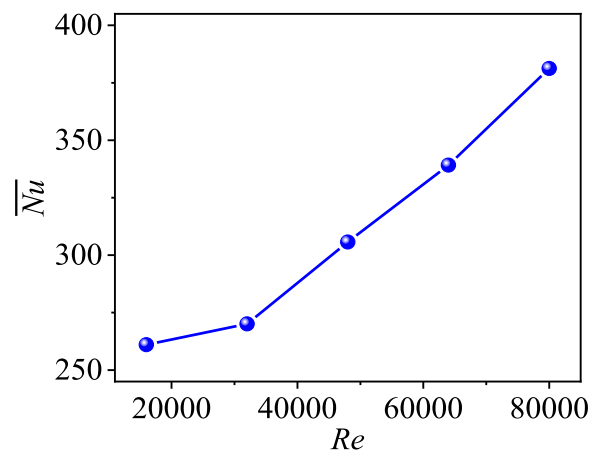
(b) Temperature uniformity index

Fig. 13. Effect of inlet flow rate on the temperature characteristics ($Ste = 0.032$, $P = 270$ W)

(a) CPU temperature augment (b) Temperature uniformity index.



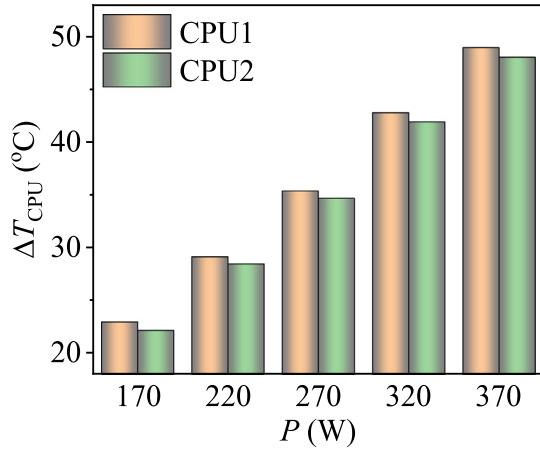
(a) Flow resistance loss



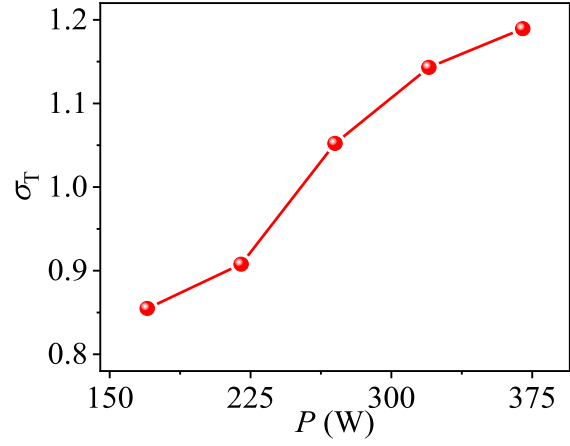
(b) Average Nusselt number

Fig. 14. Effect of inlet flow rate on the flow and thermal characteristics ($Ste = 0.032$, $P = 270$ W)

(a) Flow resistance loss (b) Average Nusselt number.

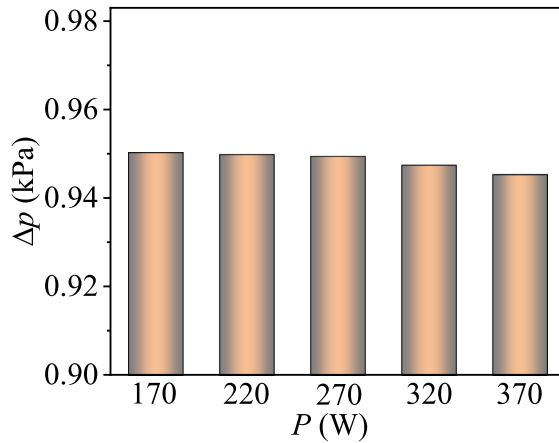


(a) CPU temperature augment

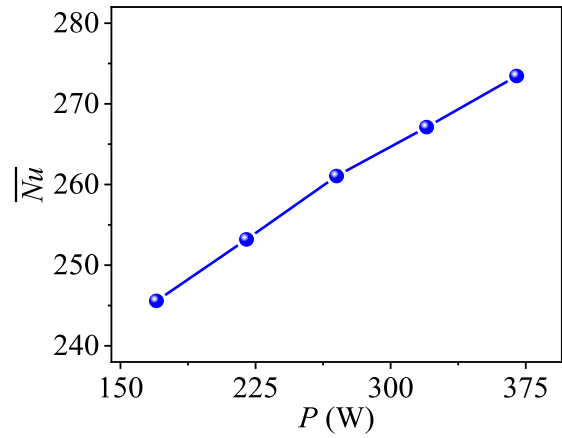


(b) Temperature uniformity index

Fig. 15. Effect of CPU power on the temperature characteristics ($Re = 16,000$, $Ste = 0.032$)
(a) CPU temperature augment (b) Temperature uniformity index.



(a) Flow resistance loss



(b) Average Nusselt number

Fig. 16. Effect of CPU power on the flow and thermal characteristics ($Re = 16,000$, $Ste = 0.032$)
(a) Flow resistance loss (b) Average Nusselt number.

the inlet Re follows the trend of an exponential relationship. Moreover, the flow resistance inside the pump-driven immersion cooling tank becomes larger as the inlet Re increases. This is mainly due to the fact that as the inlet Re increases, the flow disturbance inside the pump-driven immersion cooling tank also increases, and thus the resulting flow resistance loss also increases accordingly. In addition, Fig. 14(b) shows that the average Nusselt number in pump-driven immersion cooling tanks increases gradually with the increase of Re . According to the statistics, the corresponding average Nusselt number increases by 46.0% as Re increases from 16,000 to 80,000. As mentioned above, this is precisely due to the increase in convective perturbation with increasing Re , which intensifies the heat exchange between the fluorinert electronic liquids and server veneer electronics. In conclusion, although the increase of Re can strengthen the forced convection heat exchange and improve the heat transfer uniformity, it also increases the flow resistance loss, which in turn increases the cooling energy consumption of pump-driven immersion cooling systems.

3.2.3. Effect of CPU power

Fig. 15 shows the CPU temperature augment and overall temperature uniformity index for pump-driven immersion cooling

tanks with different CPU powers, which illustrates the influence of the CPU power on immersion cooling performance. As illustrated in Fig. 15(a), the maximum temperature augment of CPUs shows a linear increase with increasing CPU power. Moreover, the evolution trend of maximum temperature augments of CPU1 and CPU2 is almost the same, which is similar to the influence of Ste mentioned above. However, the natural convection intensity increases as the power consumption of CPUs increases, leading to the enhanced disturbance in pump-driven immersion cooling tanks, which allows for a reduction in the internal heat transfer uniformity. This conjecture is verified in the variation curve of the temperature uniformity index with CPU power shown in Fig. 15(b). Statistically speaking, the temperature uniformity index inside pump-driven immersion cooling tanks rises by 39.1% as the CPU power increases from 270 to 370 W. This phenomenon shows that the increase in CPU power is not conducive to the improvement in heat transfer uniformity inside the pump-driven immersion cooling tank.

Fig. 16 plots the relationship between the flow resistance loss, average Nusselt number and CPU power consumption. As can be observed from Fig. 16(a), the pressure drop between the inlet and outlet of pump-driven immersion cooling tanks decreases by only 0.5% with the increased CPU power, which is almost unchanged.

This phenomenon indicates that the local natural convection enhancement due to the increase in CPU power consumption has almost no effect on the pressure drop of pump-driven immersion cooling tanks. Moreover, it is evident from Fig. 16(b) that the CPU power is positively correlated with the average Nusselt number, which increases almost linearly as the CPU power increases. As the CPU power increases from 270 to 370 W, the corresponding average Nusselt number increases by 11.3%. This is due to the fact that the thermal transport temperature difference between the CPU and fluorinert electronic liquids rises as the CPU power increases, which enhances the potential for forced convection heat transfer, resulting in improved performance of the pump-driven immersion cooling tank.

4. Conclusion

To address the challenges of high energy consumption brought by the development trend of highly dense, integrated, and intelligent data centers, this work focuses on the SPIC solution that is very promising for current applications. The steady-state 3D mathematical models of flow and thermal transport processes of pump-driven and buoyancy-driven immersion cooling tanks were established, respectively. The heat transfer characteristics, flow resistance, and energy consumption levels of immersion cooling tanks in three different operating modes were investigated to reveal the immersion cooling mechanisms of pump-driven and buoyancy-driven SPIC systems. Moreover, the influences of the volume flow rate, inlet temperature, and CPU power on immersion cooling characteristics were analyzed in depth, guiding the design and operation of SPIC systems in practical engineering. In summary, the current work yields the below conclusions.

- (1) For the same operating conditions, the heat dissipation of buoyancy-driven immersion cooling tanks is less effective than pump-driven immersion cooling tanks. Compared to the buoyancy-driven one, the average temperature of fluorinert electronic liquids and the peak temperature of CPU in pump-driven immersion cooling tanks decreases by 55.5% and 40%, respectively. Moreover, the velocity distribution is relatively uniform, and the flow disturbances range is larger in a buoyancy-driven immersion cooling tank. Significantly, the PUE of pump-driven SPIC systems is reduced by 11.6% compared to buoyancy-driven SPIC systems.
- (2) The intensity of convective heat transfer in nature convection operation is poor, and its heat dissipation efficiency is much lower than in mixed convection and forced convection operating modes. Interestingly, while the natural convection in the mixed convection operation strengthens convective heat transfer to some extent, it also leads to the temperature stratification of fluorinert electronic liquids, which weakens the heat transfer homogeneity inside immersion cooling tanks.
- (3) Increases in inlet temperature and CPU power both worsen the immersion cooling performance. However, increasing inlet temperature improves heat transfer uniformity and reduces flow resistance losses, while increasing CPU power has the opposite effect. In addition, the increased flow rate enhances the convective thermal transport performance and temperature uniformity of immersion cooling tanks, but it also increases the flow resistance losses.

This paper focuses on the flow and heat transfer mechanisms within SPIC systems and the factors that influence them. In the following work, we will carry out structural design optimization of SPIC systems and investigate in depth the intrinsic mechanisms between the thermo-physical properties of fluorinert electronic liquids and the performance of SPIC systems.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Yongping Huang: Data curation, Formal analysis, Writing – original draft. **Junlei Ge:** Writing – original draft. **Yongping Chen:** Conceptualization, Methodology, Writing – review & editing. **Chengbin Zhang:** Methodology, Writing – review & editing.

Data availability

Data will be made available on request.

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